

2022 Ph H1 Q2

Section: Our Dynamic Universe
Topic: Motion, Equations and Graphs

Question Summary:

A trolley is released from rest at point **P** and accelerates uniformly down a slope. It passes through a light gate at point **Q**, which is connected to a timer. A card is attached to the trolley, and the timer records the time the **card** takes to pass through the gate.

The aim is to find the **acceleration** of the trolley using the equation:

$v^2 = u^2 + 2as$

The student makes three statements:

- I. $u = 0\text{ m/s}$
- II. $s = \text{length of the card}$
- III. $v = \frac{\text{distance from P to Q}}{\text{time displayed on the timer}}$

Which statements are correct?

Statement I: True

- The trolley is **released from rest**, so:

$u = 0\text{ m/s}$

Statement II: False

- The equation $v^2 = u^2 + 2as$ describes motion from **P to Q**, so s must be the **distance from P to Q**, not the **length of the card**
- The **length of the card** relates to measuring **final velocity**, not displacement

Statement III: False

- The time recorded by the timer is the **duration the card is within the light gate**, not the time from P to Q
- Therefore, using:
$$v = \frac{\text{distance from P to Q}}{\text{time displayed on timer}}$$

is invalid, because the **distance and time** do not correspond to the same section of motion

 Correct Answer:

A
(Only **Statement I** is correct)

Revision Tips:

- **Kinematic equations** apply to a specific segment of motion — make sure all variables refer to the same interval
- Use the **length of the card** and its **passage time** to calculate **instantaneous velocity at Q**:
$$v = \frac{\text{length of card}}{\text{time in gate}}$$
- But when applying $v^2 = u^2 + 2as$,
 - $u = 0$
 - $v = \text{speed at Q}$
 - $s = \text{distance from P to Q}$